

3 (a) $pV / T = \text{constant}$

$$T = (6.5 \times 10^6 \times 30 \times 300) / (1.1 \times 10^5 \times 540) \quad \text{C1}$$

$$= 985 \text{ K} \quad \text{A1}$$

(b) (i) The increase in the internal energy of a system is equal to the sum of the thermal energy supplied to the system and work done on the system. B1

(ii) no thermal energy supplied to system, thus increase in internal energy = work done on system B1

so U increases B1

U increases so kinetic energy of atoms increases

and hence T increases B1

3

(c) $KE \propto T$, hence $v \propto \sqrt{T}$, and so

C1

$$\text{ratio} = \sqrt{\frac{350}{300}} = 1.1$$

A1

4 **(a)** **(i)** 0.050 m

A1